Class 14: RNASeq Mini-Project

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Table of contents

Background . . . 1
Data Import . . . 1
  Clean up (data tidying) . . . 1
DESeq Analysis . . . 2
  Setting up the DESeq object . . . 2
  Running Deseq . . . 3
  Getting results . . . 3
Volcano Plot . . . 4
Add Annotation . . . 6
  KEGG . . . 14
  GO . . . 19
  Reactome . . . 20

Background

The Data for today's mini-project comes from a knock-down study of an important HoX gene.

Data Import

```
countData <- read.csv("GSE37704_featurecounts.csv", row.names = 1)
colData <- read.csv("GSE37704_metadata.csv")
```

Clean up (data tidying)

```r
countData <- as.matrix(countData[,-1])

head(colData)
```

```
         id     condition
1 SRR493366 control_sirna
2 SRR493367 control_sirna
3 SRR493368 control_sirna
4 SRR493369      hoxa1_kd
5 SRR493370      hoxa1_kd
6 SRR493371      hoxa1_kd
```

```r
head(countData)
```

```
                SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

```r
countData <- countData[rowSums(countData) != 0, ]
head(countData)
```

```
                SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000279457        23        28        29        29        28        46
ENSG00000187634       124       123       205       207       212       258
ENSG00000188976      1637      1831      2383      1226      1326      1504
ENSG00000187961       120       153       180       236       255       357
ENSG00000187583        24        48        65        44        48        64
ENSG00000187642         4         9        16        14        16        16
```

DESeq Analysis

Setting up the DESeq object

```r
library(DESeq2)
```

Running Deseq

```r
dds = DESeqDataSetFromMatrix(countData=countData,
                             colData=colData,
                             design=~condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

Getting results

```r
dds = DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```r
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
  ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(3): id condition sizeFactor
```

```r
res = results(dds)
```

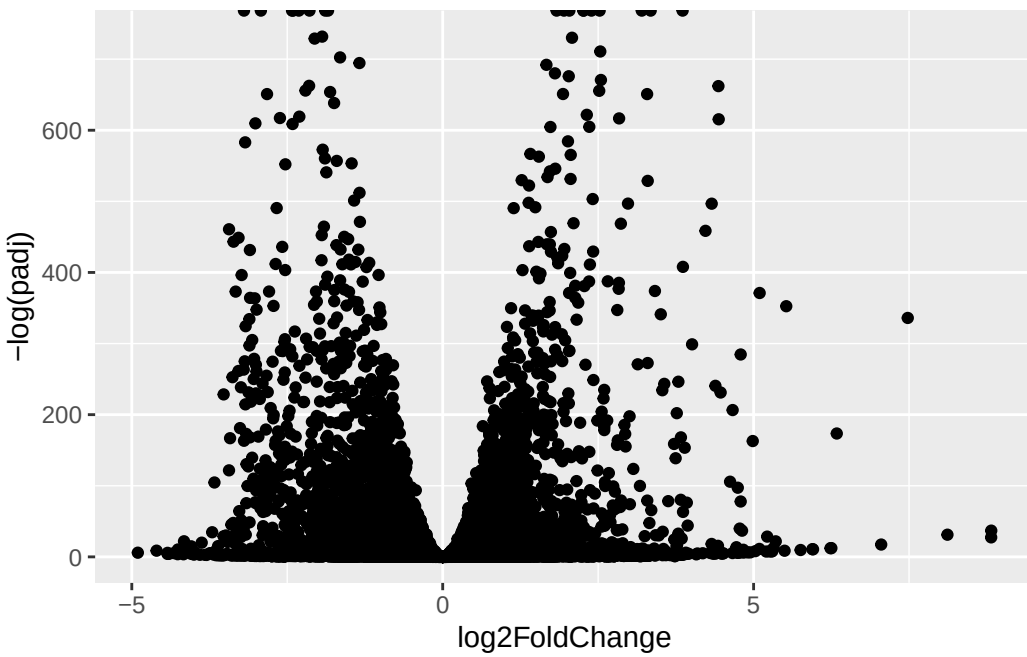
```r
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4349, 27%
LFC < 0 (down)     : 4396, 28%
outliers [1]       : 0, 0%
low counts [2]     : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano Plot

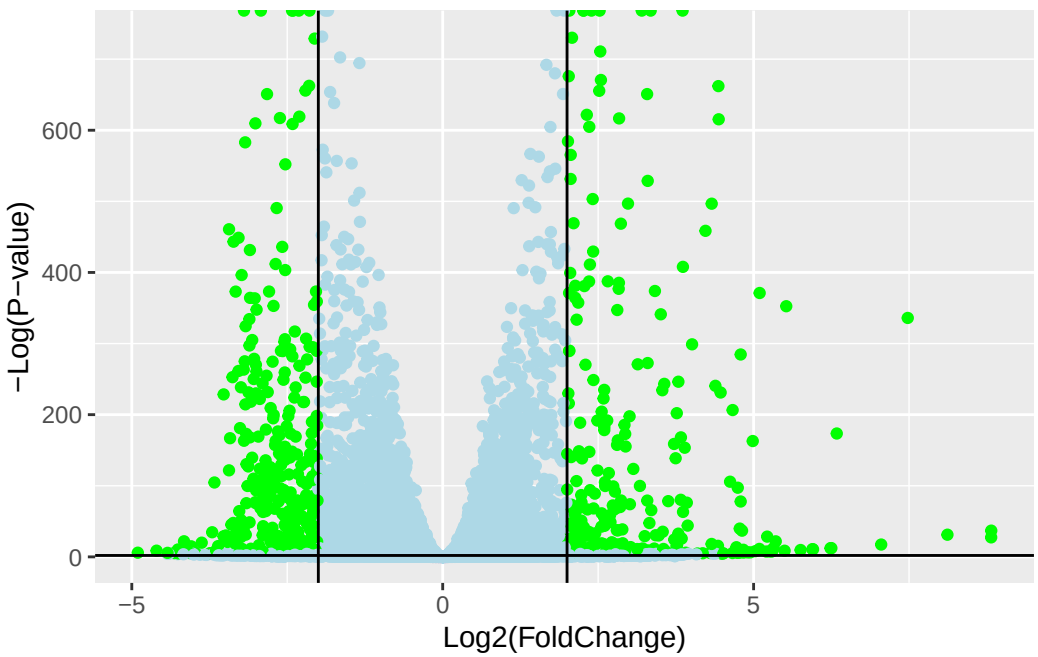
```r
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point()
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
mycols <- rep("lightblue", nrow(res) )

mycols[ abs(res$log2FoldChange) > 2 ] <- "green"

mycols[ res$padj > 0.01 ] <- "lightblue"

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(col= mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log10(0.01))
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Add Annotation

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=rownames(res),
                     keytype="ENSEMBL",
```

```
                     column="SYMBOL",
                     multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=rownames(res),
                     keytype="ENSEMBL",
                     column="ENTREZID",
                     multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$name <-  mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="GENENAME",
                     multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
head(res, 10)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 10 rows and 9 columns
                   baseMean log2FoldChange     lfcSE       stat      pvalue
                  <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.913579      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000187634  183.229650      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.188076     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.637938      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.255123      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.979750      0.5428105 0.5215598   1.040744 2.97994e-01
ENSG00000188290  108.922128      2.0570638 0.1969053  10.446970 1.51282e-25
ENSG00000187608  350.716868      0.2573837 0.1027266   2.505522 1.22271e-02
ENSG00000188157 9128.439422      0.3899088 0.0467163   8.346304 7.04321e-17
ENSG00000237330    0.158192      0.7859552 4.0804729   0.192614 8.47261e-01
                       padj      symbol      entrez                   name
```

```
                  <numeric> <character> <character>            <character>
ENSG00000279457 6.86555e-01          NA          NA                     NA
ENSG00000187634 5.15718e-03      SAMD11      148398 sterile alpha motif ..
ENSG00000188976 1.76549e-35       NOC2L       26155 NOC2 like nucleolar ..
ENSG00000187961 1.13413e-07      KLHL17      339451 kelch like family me..
ENSG00000187583 9.19031e-01     PLEKHN1       84069 pleckstrin homology ..
ENSG00000187642 4.03379e-01       PERM1       84808 PPARGC1 and ESRR ind..
ENSG00000188290 1.30538e-24        HES4       57801 hes family bHLH tran..
ENSG00000187608 2.37452e-02       ISG15        9636 ISG15 ubiquitin like..
ENSG00000188157 4.21963e-16        AGRN      375790                   agrin
ENSG00000237330          NA      RNF223      401934 ring finger protein ..
```

```
res = res[order(res$pvalue),]
write.csv(res, file="deseq_results.csv")
```

##Pathway Analysis

```
library(pathview)
library(gage)
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
     1266     54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```
head(kegg.sets.hs,5)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
```

```
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"

$`hsa01100 Metabolic pathways`
   [1] "10"        "100"       "10007"     "100137049" "10020"     "10026"
   [7] "100510686" "10063"     "10157"     "10170"     "10195"     "10201"
  [13] "10229"     "10312"     "10317"     "10327"     "10331"     "1036"
  [19] "10380"     "10390"     "1040"      "10400"     "10402"     "10423"
  [25] "10449"     "10476"     "10554"     "10555"     "10558"     "1056"
  [31] "10588"     "10606"     "10621"     "10622"     "10623"     "10632"
  [37] "10654"     "1066"      "10678"     "10682"     "10690"     "10714"
  [43] "10720"     "10768"     "10797"     "10826"     "10841"     "10855"
  [49] "10873"     "10901"     "10905"     "10941"     "10975"     "10993"
  [55] "10998"     "11019"     "11041"     "1109"      "11112"     "11128"
  [61] "1119"      "1120"      "11226"     "11227"     "11232"     "112483"
  [67] "11253"     "11282"     "11285"     "113026"    "11320"     "11343"
  [73] "113451"    "113612"    "114805"    "1152"      "1158"      "1159"
  [79] "1160"      "116285"    "117248"    "119548"    "120227"    "121278"
  [85] "122481"    "122622"    "123099"    "123745"    "123876"    "124"
  [91] "124454"    "124975"    "125"       "125061"    "125965"    "125981"
  [97] "126"       "126328"    "126792"    "127"       "127124"    "128"
 [103] "128869"    "129607"    "129642"    "130"       "130013"    "131"
 [109] "1312"      "131669"    "132"       "132158"    "1327"      "132789"
 [115] "1329"      "1337"      "1339"      "1340"      "134147"    "1345"
 [121] "1349"      "1350"      "1351"      "135152"    "1352"      "1353"
 [127] "1355"      "1371"      "1373"      "137964"    "138050"    "138429"
 [133] "139596"    "140838"    "1431"      "144193"    "144245"    "145226"
 [139] "146664"    "1491"      "15"        "1503"      "150763"    "151056"
 [145] "151531"    "1537"      "154141"    "1543"      "1544"      "1548"
 [151] "1549"      "155066"    "1551"      "1553"      "1555"      "1557"
 [157] "1558"      "1559"      "1562"      "1571"      "1573"      "157506"
 [163] "1576"      "1577"      "1579"      "158"       "1581"      "1582"
 [169] "1583"      "1584"      "1585"      "1586"      "1588"      "1589"
 [175] "159"       "1593"      "1594"      "1595"      "160287"    "1603"
 [181] "1606"      "1607"      "1608"      "160851"    "1609"      "1610"
 [187] "1621"      "162417"    "162466"    "1629"      "1633"      "1635"
 [193] "1638"      "1644"      "1650"      "166929"    "168391"    "169355"
 [199] "170712"    "171568"    "1716"      "1717"      "1718"      "1719"
 [205] "1723"      "1737"      "1738"      "1743"      "1757"      "178"
 [211] "1786"      "1787"      "1788"      "1789"      "1798"      "18"
 [217] "1806"      "1807"      "1841"      "1854"      "189"       "1890"
```

```
[223] "1892"      "191"       "192134"    "1962"      "197258"    "199857"
[229] "201595"    "2023"      "2026"      "2027"      "203"       "204"
[235] "205"       "2053"      "2058"      "210"       "211"       "212"
[241] "2131"      "2132"      "2134"      "2135"      "2137"      "216"
[247] "217"       "218"       "2180"      "2181"      "2182"      "2184"
[253] "219"       "2194"      "220"       "2203"      "221"       "221223"
[259] "221823"    "222"       "2222"      "2224"      "223"       "2235"
[265] "224"       "226"       "2271"      "22845"     "22856"     "229"
[271] "22928"     "22929"     "22934"     "22978"     "230"       "23057"
[277] "231"       "23193"     "23236"     "23305"     "23382"     "23396"
[283] "23417"     "23475"     "23483"     "23498"     "23530"     "23545"
[289] "23553"     "23556"     "2356"      "23600"     "23649"     "23761"
[295] "239"       "240"       "242"       "245972"    "245973"    "246"
[301] "246721"    "247"       "248"       "249"       "250"       "251"
[307] "2523"      "2524"      "2525"      "2526"      "2527"      "2528"
[313] "2529"      "2530"      "2531"      "253558"    "2538"      "2539"
[319] "254531"    "2548"      "256435"    "2571"      "2572"      "25796"
[325] "2581"      "2582"      "2583"      "25834"     "2584"      "2585"
[331] "2588"      "25885"     "2589"      "2590"      "25902"     "2591"
[337] "2592"      "259230"    "2593"      "259307"    "2595"      "2597"
[343] "26007"     "26035"     "2618"      "262"       "26227"     "26229"
[349] "26275"     "26279"     "2628"      "26289"     "2629"      "26290"
[355] "26301"     "2632"      "26330"     "2639"      "2643"      "2645"
[361] "2650"      "2651"      "2673"      "2678"      "2683"      "2686"
[367] "2687"      "270"       "27010"     "27034"     "27087"     "27089"
[373] "27090"     "271"       "2710"      "2712"      "27124"     "27165"
[379] "272"       "2720"      "27235"     "2729"      "2730"      "27306"
[385] "2731"      "27349"     "27430"     "2744"      "2746"      "2747"
[391] "275"       "2752"      "276"       "2762"      "277"       "278"
[397] "279"       "2799"      "28"        "280"       "2805"      "2806"
[403] "2821"      "283208"    "283871"    "284098"    "284541"    "2875"
[409] "290"       "29071"     "2937"      "2954"      "29796"     "2987"
[415] "29880"     "2990"      "29906"     "29920"     "29922"     "29925"
[421] "29926"     "29929"     "29947"     "29958"     "29968"     "30"
[427] "3028"      "3030"      "3032"      "3033"      "3034"      "3067"
[433] "3073"      "3074"      "3081"      "30814"     "30815"     "30833"
[439] "30834"     "3098"      "3099"      "31"        "3101"      "314"
[445] "3141"      "3145"      "3155"      "3156"      "3157"      "3158"
[451] "316"       "317749"    "32"        "3242"      "3251"      "326625"
[457] "3283"      "3284"      "3290"      "3291"      "3292"      "3293"
[463] "3294"      "3295"      "33"        "3340"      "3373"      "337876"
[469] "339221"    "34"        "340485"    "341392"    "3417"      "3418"
[475] "3419"      "341947"    "3420"      "3421"      "3422"      "3423"
```

```
[481] "3425"      "348158"    "349565"    "35"        "353"       "36"
[487] "3612"      "3613"      "3614"      "3615"      "3620"      "3628"
[493] "3631"      "3632"      "3633"      "3636"      "37"        "3703"
[499] "3704"      "3705"      "3706"      "3707"      "3712"      "374291"
[505] "374378"    "3795"      "38"        "383"       "384"       "387787"
[511] "39"        "3906"      "391013"    "3938"      "3939"      "3945"
[517] "3948"      "3990"      "4047"      "4048"      "4051"      "4056"
[523] "411"       "4121"      "4122"      "4124"      "4128"      "4129"
[529] "4143"      "4144"      "4190"      "4191"      "4199"      "4245"
[535] "4247"      "4248"      "4249"      "427"       "4329"      "435"
[541] "4351"      "4357"      "438"       "440"       "440138"    "440567"
[547] "441024"    "441531"    "442117"    "445"       "4507"      "4508"
[553] "4509"      "4512"      "4513"      "4514"      "4519"      "4522"
[559] "4524"      "4535"      "4536"      "4537"      "4538"      "4539"
[565] "4540"      "4541"      "4548"      "4594"      "4597"      "4598"
[571] "4669"      "4694"      "4695"      "4696"      "4697"      "4698"
[577] "47"        "4700"      "4701"      "4702"      "4704"      "4705"
[583] "4706"      "4707"      "4708"      "4709"      "471"       "4710"
[589] "4711"      "4712"      "4713"      "4714"      "4715"      "4716"
[595] "4717"      "4718"      "4719"      "4720"      "4722"      "4723"
[601] "4724"      "4725"      "4726"      "4728"      "4729"      "4731"
[607] "48"        "4830"      "4831"      "4832"      "4833"      "4837"
[613] "4842"      "4843"      "4846"      "4860"      "4907"      "493911"
[619] "4942"      "4952"      "4953"      "4967"      "498"       "50"
[625] "5009"      "501"       "5033"      "5048"      "50484"     "50487"
[631] "5049"      "5050"      "5051"      "5053"      "506"       "50614"
[637] "50617"     "50700"     "50814"     "509"       "5091"      "5095"
[643] "5096"      "51"        "51004"     "5105"      "51056"     "5106"
[649] "51074"     "51082"     "51084"     "51102"     "51109"     "51144"
[655] "51166"     "51179"     "51181"     "51196"     "51227"     "51251"
[661] "51268"     "513"       "5130"      "51301"     "51380"     "51382"
[667] "514"       "51477"     "51478"     "515"       "51540"     "516"
[673] "5160"      "51601"     "51604"     "51606"     "5161"      "5162"
[679] "5167"      "5169"      "517"       "51703"     "51727"     "51728"
[685] "51733"     "51763"     "518"       "51805"     "51809"     "5198"
[691] "521"       "5211"      "5213"      "5214"      "522"       "5223"
[697] "5224"      "5226"      "523"       "5230"      "5232"      "5236"
[703] "525"       "526"       "527"       "5277"      "5279"      "528"
[709] "5281"      "5283"      "5286"      "5287"      "5288"      "5289"
[715] "529"       "5297"      "5298"      "5313"      "5315"      "5319"
[721] "5320"      "5321"      "5322"      "533"       "5330"      "5331"
[727] "5332"      "5333"      "5335"      "53354"     "5336"      "5337"
[733] "5338"      "534"       "535"       "53630"     "537"       "5372"
```

```
[739] "5373"      "539"       "53947"     "5406"      "5407"      "5408"
[745] "5409"      "54107"     "54187"     "5422"      "5424"      "5425"
[751] "5426"      "5427"      "5428"      "5430"      "5431"      "5432"
[757] "5433"      "5434"      "54344"     "5435"      "5436"      "54363"
[763] "5437"      "5438"      "5439"      "5440"      "5441"      "5444"
[769] "5445"      "5446"      "54480"     "54490"     "54575"     "54576"
[775] "54577"     "54578"     "54579"     "54600"     "54657"     "54658"
[781] "54659"     "54675"     "5471"      "54802"     "548596"    "548644"
[787] "549"       "54947"     "54963"     "54965"     "5498"      "54988"
[793] "54995"     "55163"     "55191"     "55224"     "55229"     "55256"
[799] "55276"     "55300"     "55301"     "55304"     "55312"     "55361"
[805] "5538"      "55454"     "55500"     "55512"     "55568"     "5557"
[811] "5558"      "55627"     "55650"     "55703"     "55750"     "55753"
[817] "55790"     "55808"     "55821"     "55902"     "55907"     "56052"
[823] "5625"      "56267"     "5631"      "5634"      "56474"     "56623"
[829] "56624"     "56655"     "56848"     "56894"     "56895"     "56898"
[835] "56901"     "56913"     "56922"     "56953"     "56994"     "570"
[841] "57016"     "57026"     "57134"     "5723"      "5730"      "5740"
[847] "5742"      "5743"      "57452"     "574537"    "57678"     "57804"
[853] "57818"     "57834"     "5805"      "5831"      "5832"      "5833"
[859] "58510"     "5859"      "586"       "5860"      "587"       "593"
[865] "594"       "5980"      "60490"     "60495"     "6120"      "6184"
[871] "6185"      "622"       "6240"      "6241"      "6296"      "6303"
[877] "6307"      "6309"      "6342"      "635"       "6389"      "6390"
[883] "6391"      "63917"     "6392"      "64087"     "64131"     "64132"
[889] "64409"     "64425"     "6448"      "64579"     "64600"     "646625"
[895] "6470"      "6472"      "6476"      "64768"     "6480"      "64802"
[901] "64816"     "6482"      "6483"      "6484"      "6487"      "6489"
[907] "64902"     "65220"     "65263"     "654364"    "6609"      "661"
[913] "6610"      "6611"      "6652"      "6675"      "6677"      "669"
[919] "6697"      "6713"      "6718"      "6723"      "683"       "686"
[925] "6888"      "6898"      "6916"      "6999"      "7054"      "7083"
[931] "7084"      "7086"      "7108"      "7166"      "7167"      "7173"
[937] "7263"      "7264"      "729020"    "7298"      "7299"      "7306"
[943] "7357"      "7358"      "7360"      "7363"      "7364"      "7365"
[949] "7366"      "7367"      "7368"      "7371"      "7372"      "7378"
[955] "7381"      "7384"      "7385"      "7386"      "7388"      "7389"
[961] "7390"      "7498"      "7841"      "790"       "79053"     "79087"
[967] "7915"      "79178"     "7923"      "79369"     "7941"      "79586"
[973] "79611"     "79623"     "79646"     "79695"     "79717"     "79796"
[979] "79799"     "79814"     "79868"     "79888"     "7991"      "80025"
[985] "80055"     "80142"     "80146"     "80201"     "80270"     "80308"
[991] "80339"     "80347"     "8050"      "81490"     "81579"     "81616"
```

```
 [997] "81849"     "81888"     "8277"      "8309"      "8310"      "83440"
[1003] "83549"     "8372"      "8382"      "8394"      "8395"      "8398"
[1009] "8399"      "84002"     "84076"     "84172"     "84245"     "84265"
[1015] "84274"     "84284"     "84532"     "84618"     "84620"     "84647"
[1021] "84649"     "84693"     "847"       "84701"     "84706"     "84720"
[1027] "84735"     "84803"     "84812"     "84890"     "84920"     "84992"
[1033] "8509"      "8513"      "8525"      "8526"      "8527"      "8529"
[1039] "85365"     "8540"      "85465"     "8560"      "8564"      "8566"
[1045] "8608"      "8611"      "8612"      "8613"      "8630"      "8639"
[1051] "8659"      "8681"      "8692"      "8693"      "8694"      "8702"
[1057] "8703"      "8704"      "8705"      "8706"      "8707"      "8708"
[1063] "873"       "8733"      "874"       "875"       "8760"      "8789"
[1069] "8790"      "8801"      "8802"      "8803"      "8813"      "8818"
[1075] "8821"      "883"       "8833"      "8854"      "8867"      "8869"
[1081] "8871"      "8877"      "8879"      "8942"      "8972"      "8974"
[1087] "89869"     "8992"      "9"         "90423"     "9060"      "9061"
[1093] "9091"      "9114"      "91373"     "9162"      "91734"     "9197"
[1099] "9245"      "92483"     "9249"      "9296"      "93034"     "93183"
[1105] "9331"      "9348"      "9374"      "9377"      "9380"      "9388"
[1111] "94005"     "9453"      "9468"      "9487"      "9488"      "9489"
[1117] "95"        "9514"      "9517"      "952"       "9533"      "9536"
[1123] "9550"      "9551"      "9563"      "9588"      "9615"      "978"
[1129] "9791"      "9942"      "9945"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
```

```
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
[161] "9583"   "9615"

$`hsa05340 Primary immunodeficiency`
 [1] "100"    "115650" "23495"  "29760"  "29851"  "326"    "3543"   "3561"
 [9] "3575"   "3718"   "3932"   "4261"   "57379"  "5788"   "5896"   "5897"
[17] "5993"   "5994"   "64421"  "6890"   "6891"   "695"    "7374"   "7535"
[25] "8517"   "8625"   "915"    "916"    "920"    "925"    "926"    "930"
[33] "958"    "959"    "973"
```

```
keggres <- gage(foldchanges, gsets=kegg.sets.hs)
head(keggres$less)
```

```
                                                   p.geomean stat.mean
hsa04110 Cell cycle                             8.995727e-06 -4.378644
hsa03030 DNA replication                        9.424076e-05 -3.951803
hsa05130 Pathogenic Escherichia coli infection 1.405864e-04 -3.765330
hsa03013 RNA transport                          1.375901e-03 -3.028500
hsa03440 Homologous recombination               3.066756e-03 -2.852899
hsa04114 Oocyte meiosis                         3.784520e-03 -2.698128
                                                       p.val       q.val
hsa04110 Cell cycle                             8.995727e-06 0.001889103
hsa03030 DNA replication                        9.424076e-05 0.009841047
hsa05130 Pathogenic Escherichia coli infection 1.405864e-04 0.009841047
hsa03013 RNA transport                          1.375901e-03 0.072234819
hsa03440 Homologous recombination               3.066756e-03 0.128803765
hsa04114 Oocyte meiosis                         3.784520e-03 0.132458191
                                                set.size         exp1
hsa04110 Cell cycle                                  121 8.995727e-06
hsa03030 DNA replication                              36 9.424076e-05
hsa05130 Pathogenic Escherichia coli infection        53 1.405864e-04
hsa03013 RNA transport                               144 1.375901e-03
hsa03440 Homologous recombination                     28 3.066756e-03
hsa04114 Oocyte meiosis                              102 3.784520e-03
```

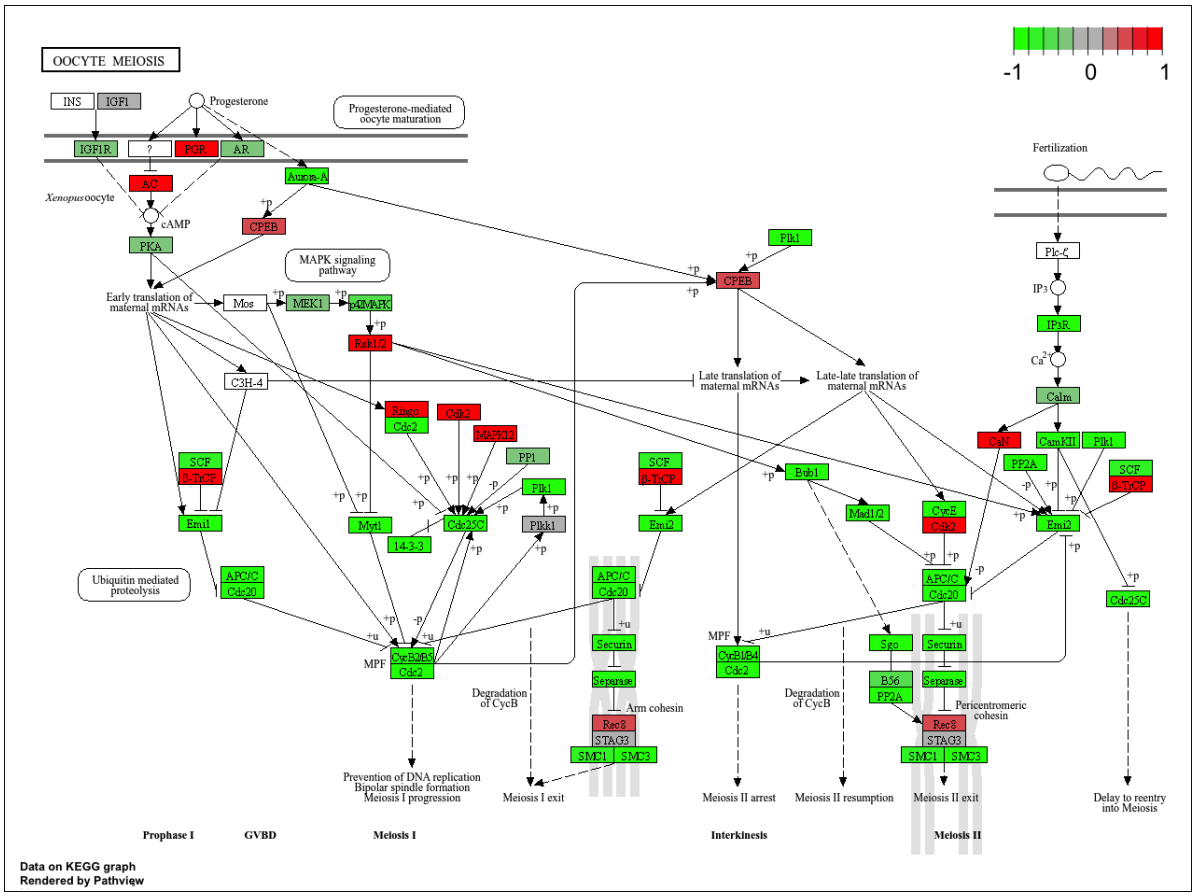
KEGG

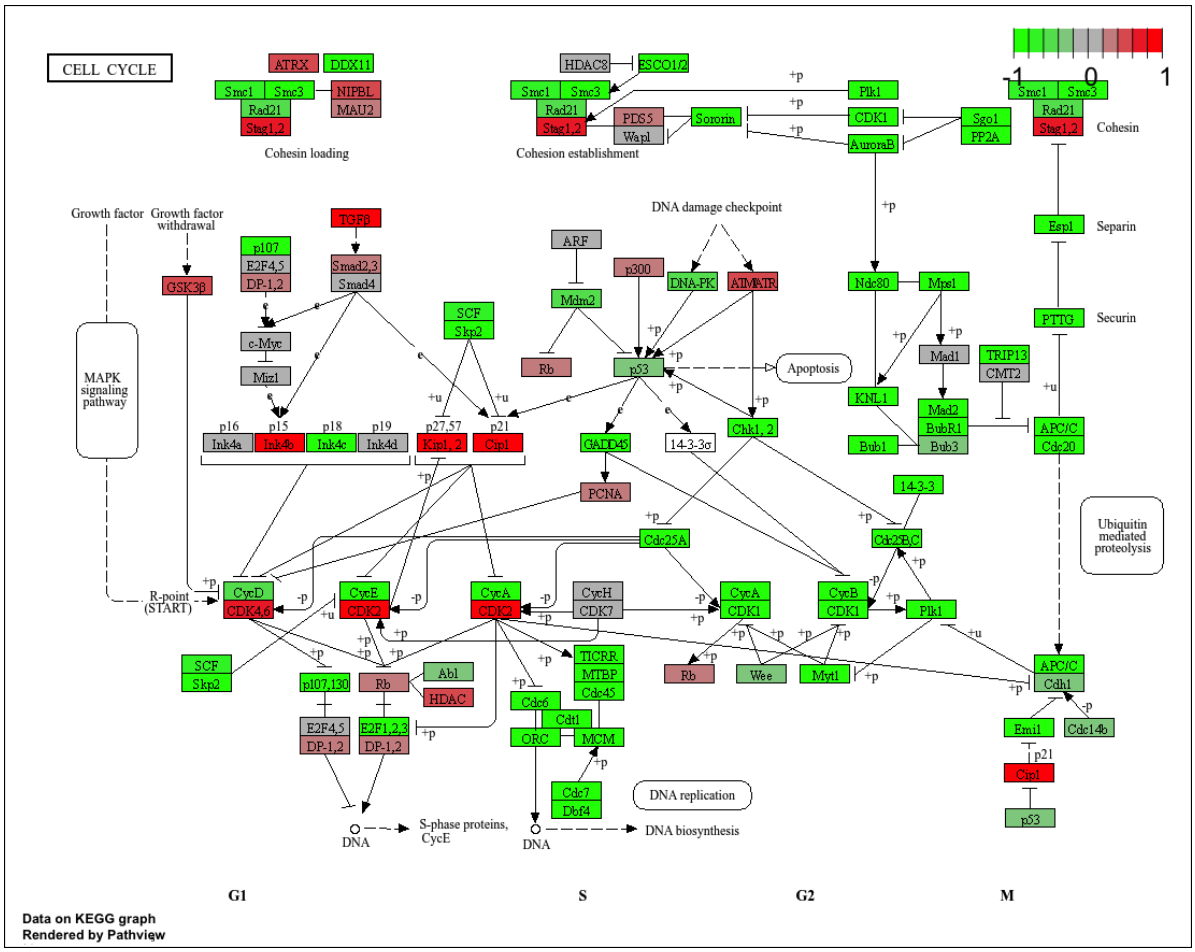
```
keggrespathways <- rownames(keggres$less)[1:5]
```

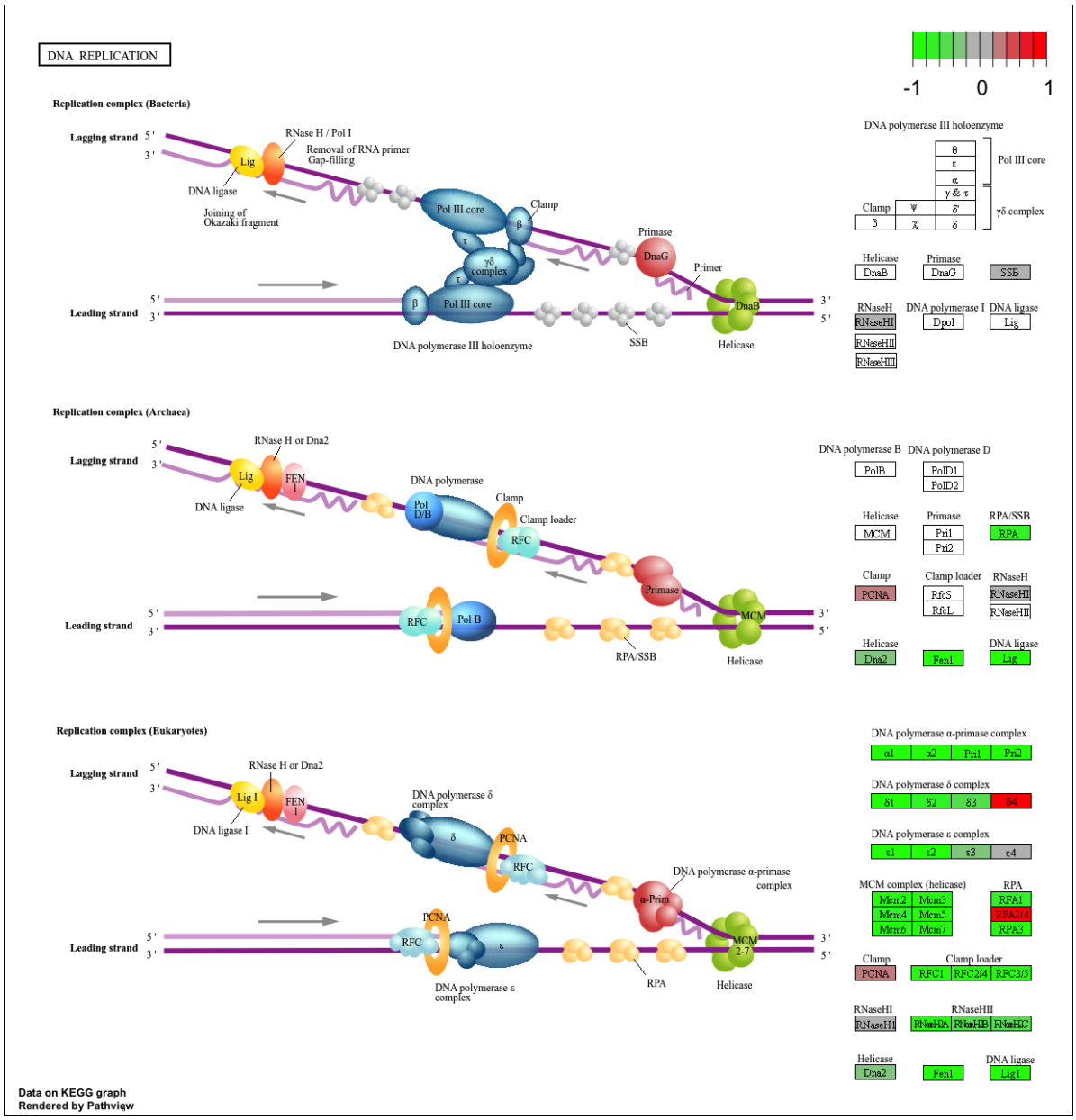
```r
keggresids <- substr(keggrespathways, start=1, stop=8)

keggresids
```

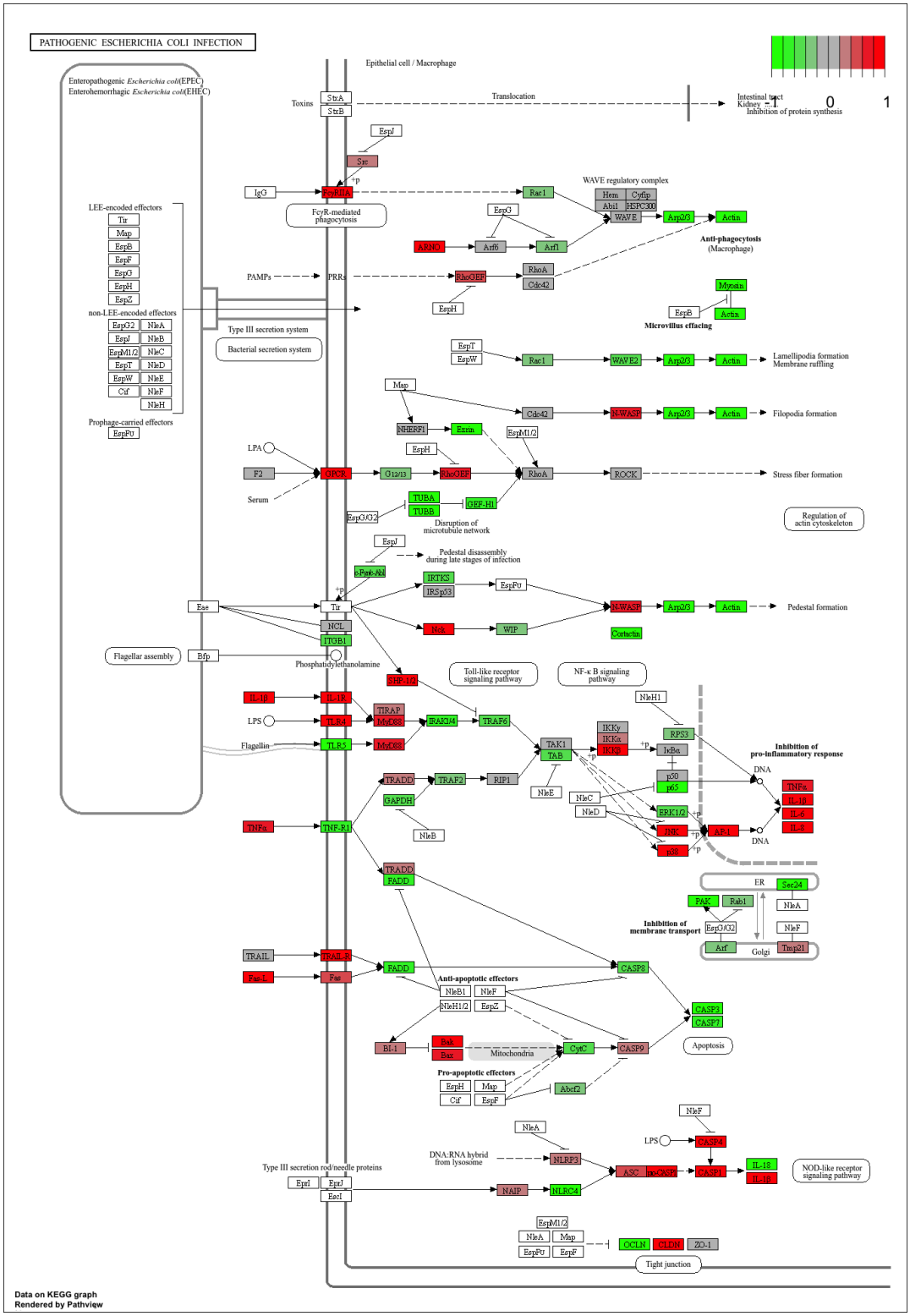
```
[1] "hsa04110" "hsa03030" "hsa05130" "hsa03013" "hsa03440"
```

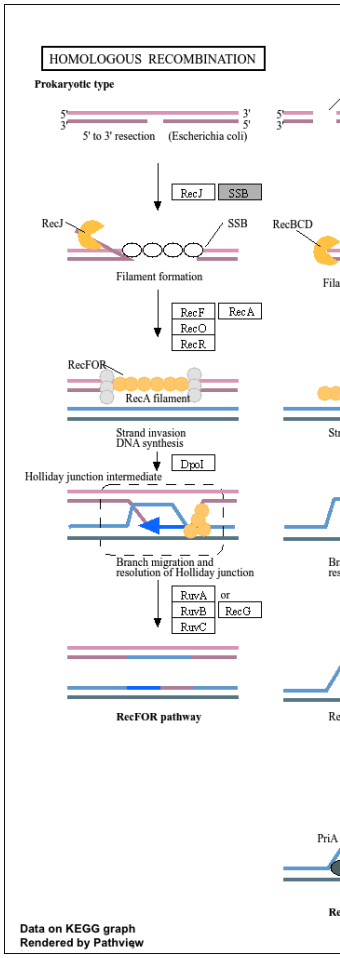
```r
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```




CELL CYCLE
-1
0
1
ATRX
DDX11
Smc1
Smc3
NIPBL
MAU2
Rad21
Stag1,2
Cohesin loading
HDAC8
ESCO1/2
Smc1
Smc3
Rad21
Stag1,2
PDS5
Wapl
Sororin
Cohesion establishment
Plk1
CDK1
AuroraB
Sgo1
PP2A
Smc1
Smc3
Rad21
Stag1,2
Cohesin
Espl
Separin
PTTG
Securin
Growth factor
Growth factor withdrawal
TGFβ
Smad2,3
Smad4
p107
E2F4,5
DP-1,2
GSK3β
c-Myc
Miz1
MAPK signaling pathway
SCF
Skp2
p16
Ink4a
p15
Ink4b
p18
Ink4c
p19
Ink4d
p27,57
Kip1, 2
p21
Cip1
ARF
Mdm2
Rb
p300
DNA-PK
ATM/ATR
DNA damage checkpoint
p53
Apoptosis
GADD45
14-3-3σ
Chk1, 2
PCNA
Cdc25A
Ndc80
Mps1
KNL1
Mad1
TRIP13
CMT2
Mad2
BubR1
Bub1
Bub3
APC/C
Cdc20
14-3-3
Cdc25B,C
Ubiquitin mediated proteolysis
R-point (START)
CycD
CDK4,6
CycE
CDK2
CycA
CDK2
CycH
CDK7
CycA
CDK1
CycB
CDK1
Plk1
SCF
Skp2
p107,130
Rb
Abl
HDAC
TICRR
MTBP
Cdc45
Rb
Wee
Myt1
APC/C
Cdh1
Cdc6
Cdt1
ORC
MCM
E2F4,5
DP-1,2
E2F1,2,3
DP-1,2
Emi1
Cdc14b
p21
Cip1
p53
Cdc7
Dbf4
DNA replication
DNA
S-phase proteins, CycE
DNA
DNA biosynthesis
G1
S
G2
M
Data on KEGG graph
Rendered by Pathview


DNA REPLICATION
-1
0
1
Replication complex (Bacteria)
Lagging strand
5'
3'
RNase H / Pol I
Removal of RNA primer
Gap-filling
Lig
DNA ligase
Joining of
Okazaki fragment
Pol III core
Clamp
β
τ
γδ
complex
Primase
DnaG
Primer
Leading strand
DNA polymerase III holoenzyme
SSB
DnaB
Helicase
DNA polymerase III holoenzyme
θ
ε
α
γ & τ
Pol III core
Clamp
ψ
δ'
γδ complex
β
χ
δ
Helicase
DnaB
Primase
DnaG
SSB
RNaseH
RNaseHI
RNaseHII
RNaseHIII
DNA polymerase I
DpoI
DNA ligase
Lig
Replication complex (Archaea)
RNase H or Dna2
FEN
1
DNA polymerase
Pol
D/B
Clamp loader
RFC
Pol B
MCM
RPA/SSB
DNA polymerase B
PolB
DNA polymerase D
PolD1
PolD2
Helicase
MCM
Primase
Pri1
Pri2
RPA/SSB
RPA
Clamp
PCNA
Clamp loader
RfcS
RfcL
RNaseH
RNaseHI
RNaseHIII
Helicase
Dna2
Fen1
DNA ligase
Lig
Replication complex (Eukaryotes)
RNase H or Dna2
Lig I
FEN
DNA ligase I
DNA polymerase δ
complex
δ
PCNA
RFC
DNA polymerase α-primase
complex
α-Prim
ε
DNA polymerase ε
complex
MCM
2-7
RPA
Helicase
DNA polymerase α-primase complex
α1
α2
Pri1
Pri2
DNA polymerase δ complex
δ1
δ2
δ3
δ4
DNA polymerase ε complex
ε1
ε2
ε3
ε4
MCM complex (helicase)
Mcm2
Mcm3
Mcm4
Mcm5
Mcm6
Mcm7
RPA
RFA1
RFA2/4
RFA3
Clamp
PCNA
Clamp loader
RFC1
RFC2/4
RFC3/5
RNaseHI
RNaseHI1
RNaseHIII
RNaseHIA
RNaseHIB
RNaseHIC
Helicase
Dna2
Fen1
DNA ligase
Lig1
Data on KEGG graph
Rendered by Pathview


PATHOGENIC ESCHERICHIA COLI INFECTION
Epithelial cell / Macrophage
Enteropathogenic Escherichia coli(EPEC)
Enterohemorrhagic Escherichia coli(EHEC)
Toxins
StxA
StxB
Translocation
Intestinal tract
Kidney
Inhibition of protein synthesis
0
1
EspJ
Src
IgG
FcγR-mediated phagocytosis
Rac1
WAVE regulatory complex
Hem
Cyfip
Abi1
HSPC300
WAVE
Arp2/3
Actin
Anti-phagocytosis
(Macrophage)
EspG
ARNO
Arf6
Arf1
LEE-encoded effectors
Tir
Map
EspB
EspF
EspG
EspH
EspZ
non-LEE-encoded effectors
EspG2
NleA
EspJ
NleB
EspM1/2
NleC
EspT
NleD
EspW
NleE
Cif
NleF
NleH
Prophage-carried effectors
EspFu
PAMPs
PRRs
RhoGEF
RhoA
Cdc42
EspH
Myosin
EspB
Actin
Microvillus effacing
Type III secretion system
Bacterial secretion system
EspT
EspW
Rac1
WAVE2
Arp2/3
Actin
Lamellipodia formation
Membrane ruffling
Map
Cdc42
N-WASP
Arp2/3
Actin
Filopodia formation
NHERF1
Ezrin
EspM1/2
EspH
LPA
F2
GPCR
G12/13
RhoGEF
RhoA
ROCK
Stress fiber formation
Serum
TUBA
TUBB
GEF-H1
EspG/G2
Disruption of microtubule network
Regulation of actin cytoskeleton
EspJ
Pedestal disassembly during late stages of infection
c-Pept-Abl
IRTKS
IRSp53
EspFu
Eae
Tir
NCL
ITGB1
N-WASP
Arp2/3
Actin
Pedestal formation
Nck
WIP
Cortactin
Flagellar assembly
BfpD
Phosphatidylethanolamine
SHP-1/2
Toll-like receptor signaling pathway
NF-κ B signaling pathway
NleH1
IL-1β
IL-1R
LPS
TLR4
TRAP
MyD88
IRAK1/4
TRAF6
Flagellin
TLR5
MyD88
TAK1
TAB
IKKγ
IKKα
IKKβ
RPS3
IκBα
Inhibition of pro-inflammatory response
DNA
TRADD
TRAF2
RIP1
GAPDH
NleE
NleC
NleD
p50
p65
ERK1/2
JNK
p38
AP-1
DNA
TNFα
IL-1β
IL-6
IL-8
TNFα
TNF-R1
NleB
TRADD
FADD
ER
Sec24
PAK
Rab1
NleA
Inhibition of membrane transport
EspG/G2
NleF
Arf
Golgi
Tmp21
TRAIL
TRAIL-R
Fas-L
Fas
FADD
Anti-apoptotic effectors
NleB1
NleF
NleH1/2
EspZ
CASP8
CASP3
CASP7
Bcl-1
Bak
Bax
Mitochondria
CytC
CASP9
Apoptosis
Pro-apoptotic effectors
EspH
Map
Cif
EspF
AbcF2
NleF
NleA
LPS
CASP4
DNA:RNA hybrid from lysosome
NLRP3
ASC
proCASP1
CASP1
IL-18
IL-1β
NOD-like receptor signaling pathway
Type III secretion rod/needle proteins
EprI
EprJ
EscI
NAIP
NLRC4
EspM1/2
NleA
Map
EspFu
EspF
OCLN
CLDN
ZO-1
Tight junction
Data on KEGG graph
Rendered by Pathview


HOMOLOGOUS RECOMBINATION
Prokaryotic type
5' to 3' resection
(Escherichia coli)
RecJ
SSB
RecJ
SSB
RecBCD
Filament formation
RecF
RecA
RecO
RecR
RecFOR
RecA filament
Strand invasion
DNA synthesis
DpoI
Holliday junction intermediate
Branch migration and resolution of Holliday junction
RuvA
or
RuvB
RecG
RuvC
RecFOR pathway
Data on KEGG graph
Rendered by Pathview

GO

```r
data(go.sets.hs)
data(go.subs.hs)

gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)
```

```
$greater
                                            p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion       8.519724e-05  3.824205 8.519724e-05
GO:0002009 morphogenesis of an epithelium 1.396681e-04  3.653886 1.396681e-04
GO:0048729 tissue morphogenesis           1.432451e-04  3.643242 1.432451e-04
GO:0007610 behavior                       1.925222e-04  3.565432 1.925222e-04
GO:0060562 epithelial tube morphogenesis  5.932837e-04  3.261376 5.932837e-04
GO:0035295 tube development               5.953254e-04  3.253665 5.953254e-04
                                              q.val set.size         exp1
GO:0007156 homophilic cell adhesion       0.1951953      113 8.519724e-05
GO:0002009 morphogenesis of an epithelium 0.1951953      339 1.396681e-04
GO:0048729 tissue morphogenesis           0.1951953      424 1.432451e-04
GO:0007610 behavior                       0.1967577      426 1.925222e-04
GO:0060562 epithelial tube morphogenesis  0.3565320      257 5.932837e-04
GO:0035295 tube development               0.3565320      391 5.953254e-04

$less
                                              p.geomean stat.mean        p.val
GO:0048285 organelle fission             1.536227e-15 -8.063910 1.536227e-15
GO:0000280 nuclear division              4.286961e-15 -7.939217 4.286961e-15
GO:0007067 mitosis                       4.286961e-15 -7.939217 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.169934e-14 -7.797496 1.169934e-14
GO:0007059 chromosome segregation        2.028624e-11 -6.878340 2.028624e-11
GO:0000236 mitotic prometaphase          1.729553e-10 -6.695966 1.729553e-10
                                                 q.val set.size         exp1
GO:0048285 organelle fission             5.841698e-12      376 1.536227e-15
GO:0000280 nuclear division              5.841698e-12      352 4.286961e-15
GO:0007067 mitosis                       5.841698e-12      352 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.195672e-11      362 1.169934e-14
GO:0007059 chromosome segregation        1.658603e-08      142 2.028624e-11
```

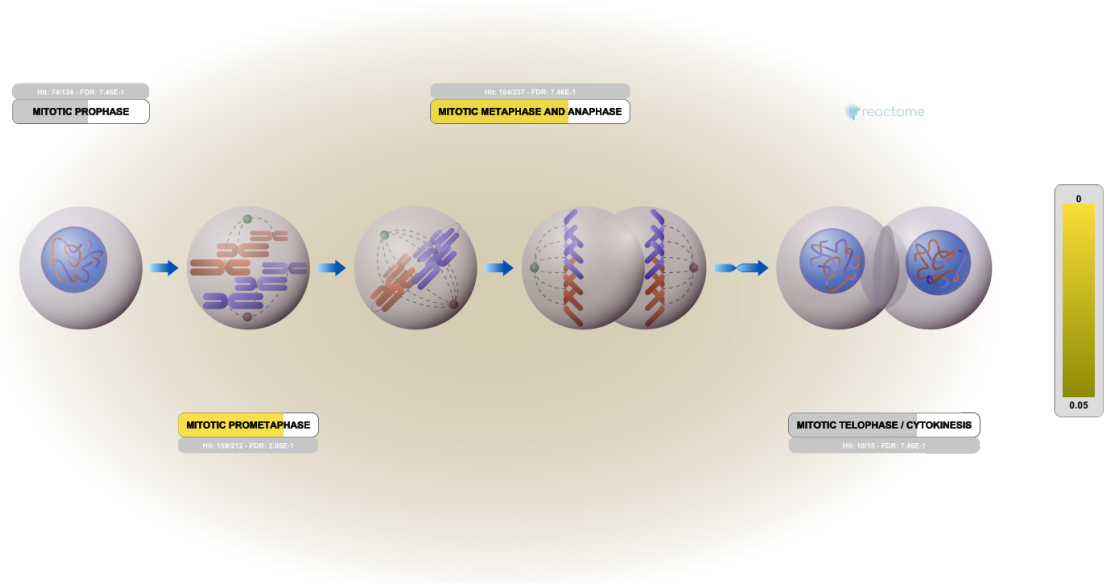
```
GO:0000236 mitotic prometaphase          1.178402e-07       84 1.729553e-10

$stats
                                          stat.mean     exp1
GO:0007156 homophilic cell adhesion        3.824205 3.824205
GO:0002009 morphogenesis of an epithelium  3.653886 3.653886
GO:0048729 tissue morphogenesis            3.643242 3.643242
GO:0007610 behavior                        3.565432 3.565432
GO:0060562 epithelial tube morphogenesis   3.261376 3.261376
GO:0035295 tube development                3.253665 3.253665
```

Reactome

```r
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```r
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```



The cell cycle have the most significant "Entities p-value" and most does not match with my KEGG results. This is most likely due to the different database used by two analysis.